

Reducing the Cost of Portability: AI-Assisted Schema Mapping for Learner Data Infrastructure

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Introductions & overview.....

DataKind builds scalable
solutions to advance
social good.

The big challenge(s)

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Postsecondary learner record data is fragmented, institution-controlled, and non-portable — limiting student agency and economic mobility.

Learner Information Framework

The Learner Information Framework (LIF) is a set of open-source tools and a shared data model that work with existing systems to find, translate, and package learner information into a consistent, governed format, so institutions, data leaders, and application teams can use it reliably and securely.

The substantial technical and financial costs associated with cross-system data mapping, transformation, and validation contribute to uneven implementation of the digital public infrastructure required for portable learner credentials.

Digital public infrastructure + AI
tooling could reduce access
barriers and accelerate adoption

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Primary research questions

1. Can AI support data quality assessment such that we have confidence in the data that is being shared?
2. Can AI support mapping from source data system to target data system in order to increase record portability?
3. Can AI support data transformation between source data system and target data system in order to ensure data is harmonized to a shared data standard?

Data quality assessment



Data Observation Toolkit (DOT)

Open-source toolkit for automated data quality monitoring

50+ data quality tests specific to education schemas



Example Tests for Assessment Schema:

- Effective date $\leq$ expiration date (when both present)
 - If `dateEffective` and `expirationDate` are both non-missing, `dateEffective <= expirationDate`.
- Deleted metadata consistency checks
 - If `deletedDate` is present, `deletedStatus` should also be present (e.g., “Deleted”, “SoftDeleted”).
 - If `deletedStatus` indicates deletion, then `deletedDate` must be populated and not in the future.

System mapping



Secondary research questions

Unlike simple text matching, AI language models (LLMs) can reason about meaning — making them a promising tool for schema mapping:

- Understand that "delivery method" and "instructional delivery" refer to the same concept
- Identify when a source field has no valid match and flag it accordingly

But key questions remain:

- How accurate is AI-assisted mapping in practice?
- Do larger, more expensive models meaningfully outperform smaller, cheaper ones?
- What is the real cost of using AI at scale?

Experimental approach

We tested 12 methods across three tiers of sophistication to find the best approach for mapping institutional data fields to the LIF data standard

Tier 1 — Traditional (4 methods): Text-matching tools that compare field names and descriptions without understanding meaning

Tier 2 — Small AI (5 methods): Compact language models that reason about semantics and follow instructions

Tier 3 — Large AI (3 methods): Frontier language models with the strongest reasoning capabilities

Experimental approach

Typical attempts to evaluate AI for schema mapping suffer from confounded variables. Multiple factors changed between runs (prompts, models, preprocessing steps), making it impossible to attribute improvements to specific changes. Was the improvement from the new model, or from the revised prompt, or from fixing a bug in preprocessing?

This experiment enforced strict variable isolation:

- When comparing models, prompts remained identical
- When comparing prompting strategies, models remained fixed
- All parameters (temperature, token limits, prompt versions) were frozen before execution
- Every run was logged with complete inputs and outputs for reproducibility

Experimental approach

- **Two source schemas:** the Postsecondary Data Partnership (PDP) schema by the National Student Clearinghouse and the Credential Transparency Description Language (CTDL) by Credential Engine
- **One target schema:** the Learner information Framework
- **Human-validated datasets** created for experimentation for acceptable matches between source and target (140 source mappings to 521 target mappings)
 - Human mapping - estimate of seven minutes per field mapped
- **Solving the “Needle-in-Haystack” challenge.** For each source field, the method must either:
 - Select the correct target from 521 options, OR Correctly determine that no valid target exists
 - This is inherently difficult. Even for fields with valid mappings, the correct answer is 1 in 521 (0.2%). Random guessing would achieve near-zero accuracy.

Experimental approach

Precision: “When the AI proposes a mapping, how often is it correct?”

- Precision=Correct proposals /Total proposals
- A precision of 70% means: of every 10 mappings the AI proposes, 7 are correct and 3 are wrong.
- *Why precision matters:* Wrong mappings that look plausible can slip through human review and cause data errors downstream. We want to minimize these false positives.

Recall: “Of all the real mappings that exist, how many did the AI find?”

- Recall = Mappings found/Total real mappings
- A recall of 80% means: if there are 30 real mappings, the AI found 24 and missed 6.
- *Why recall matters:* Missed mappings mean data doesn't transfer correctly between systems. However, missed mappings are usually obvious (a field with no mapping gets flagged for human review), so they're less dangerous than wrong mappings.

F0.5: “A combined score that weights precision more heavily than recall.”

- $F0.5 = 1.25 * (\text{Precision} * \text{Recall}) / (0.25 * \text{Precision} + \text{Recall})$
- *Why F0.5:* Standard F1 weights precision and recall equally. Since we believe wrong mappings (precision failures) are more costly than missed mappings (recall failures), we use F0.5 which weights precision twice as heavily.

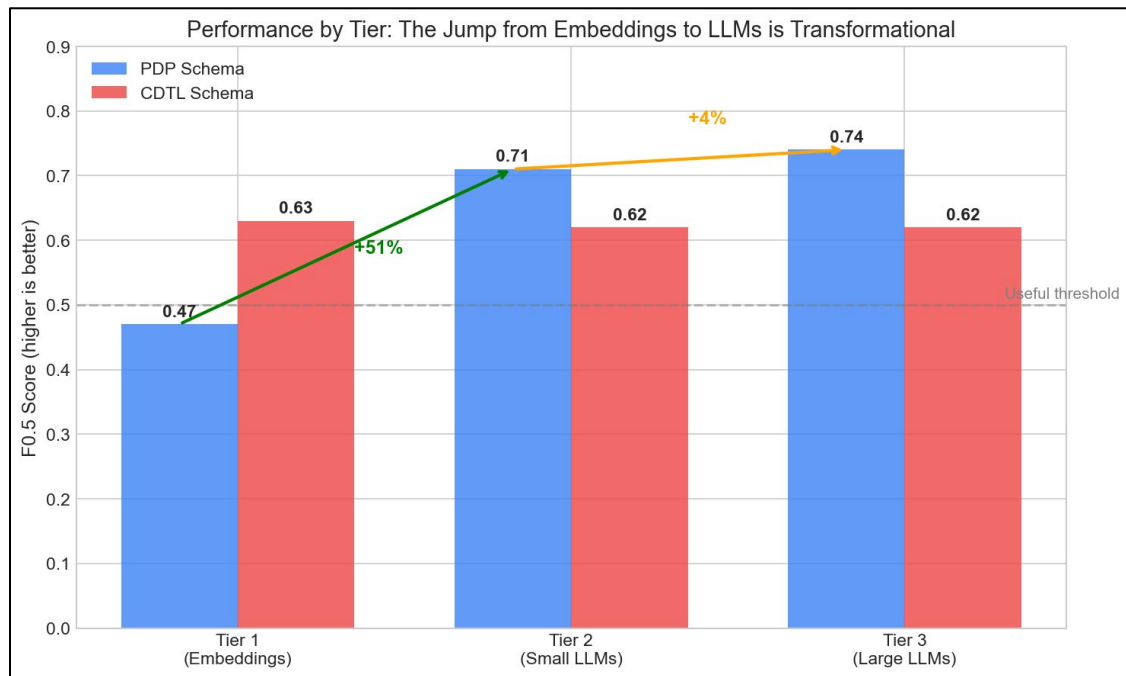
Hallucination Rate: “How often does the AI invent fields that don't exist?”

- Hallucination Rate=Predictions with non-existent targets/Total predictions
- A hallucination rate of 5% means: 5 of every 100 predictions reference a target field that isn't in the LIF schema at all. This indicates the AI is “making things up” rather than selecting from valid options.

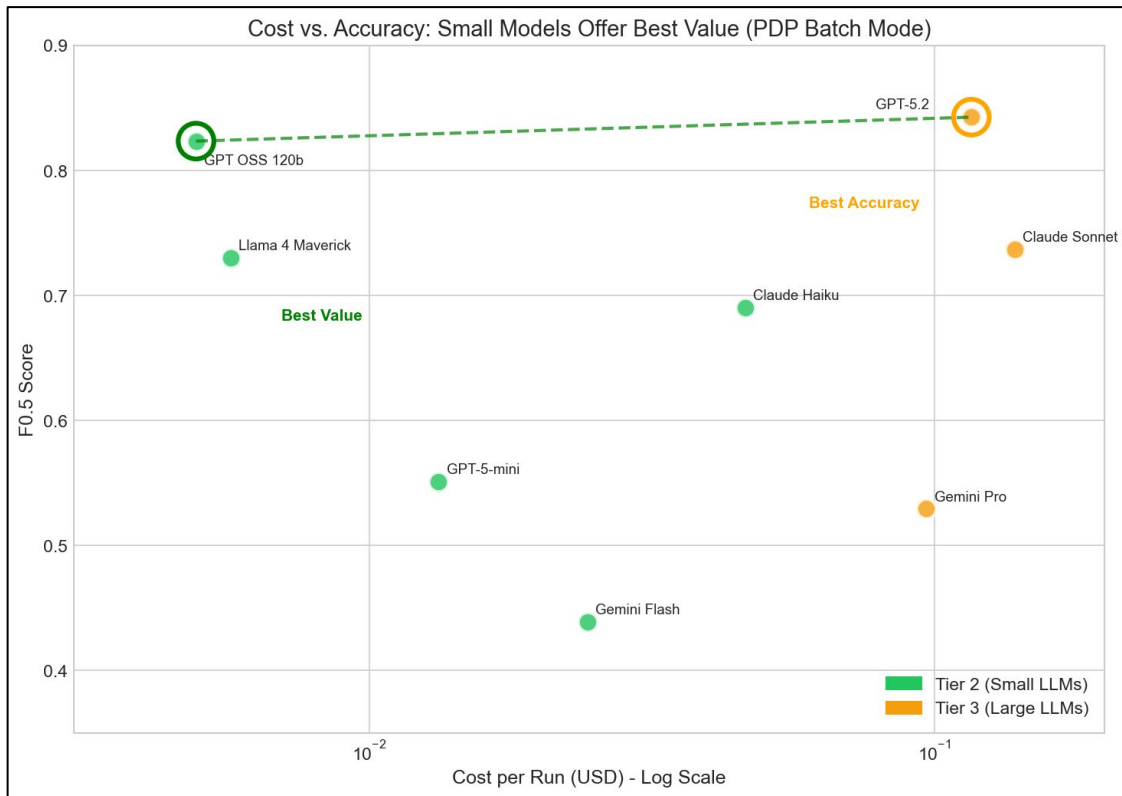
Matthews Correlation Coefficient (MCC): A single number from -1 to +1 that captures overall classification quality, accounting for class imbalance.

- MCC = +1: Perfect predictions, MCC = 0: No better than random guessing, MCC = -1: Perfectly wrong
- *Why MCC matters for this task:* Say our dataset has only 21% positive mappings (30/141). A method that always predicts “no match” would achieve 79% accuracy but 0% recall and MCC near 0. MCC correctly identifies such degenerate strategies.

Results

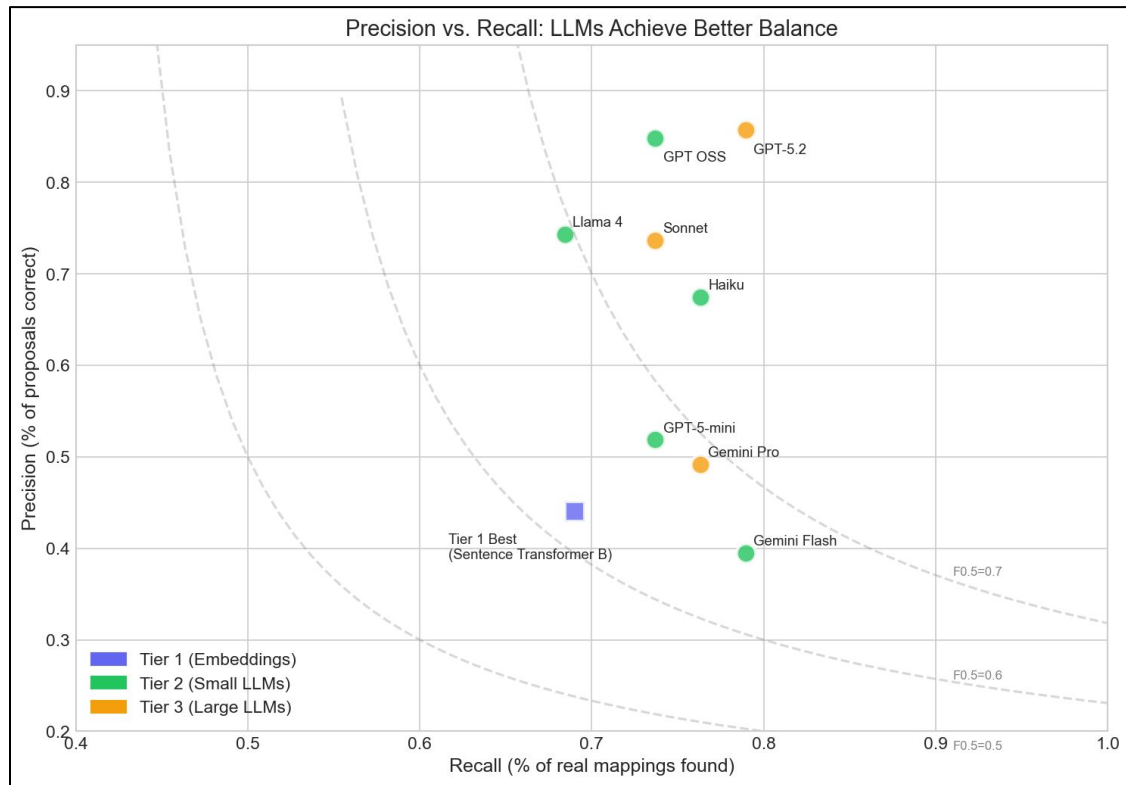


Results



Practical implication: For most use cases, the cheapest model is the best choice. The 24x cost premium for GPT-5.2 yields only 4% accuracy improvement. *Cost vs. Accuracy: Small Models Offer Best Value*

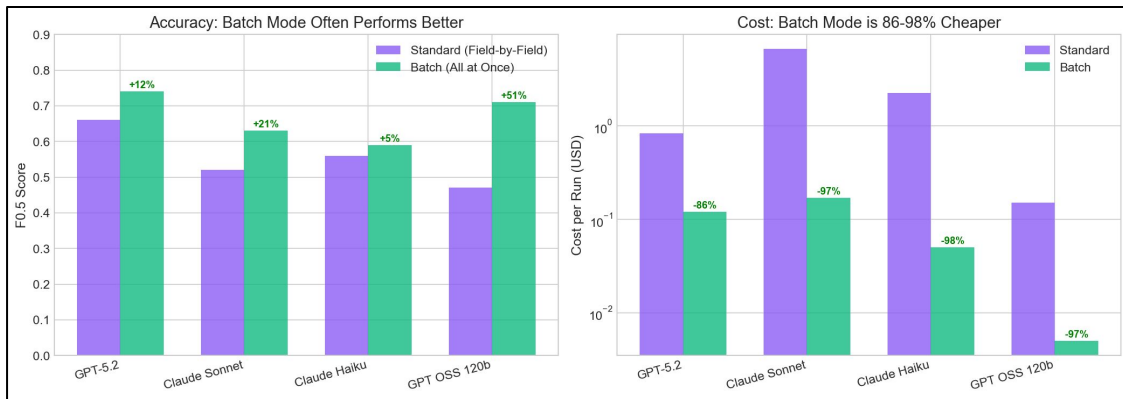
Results



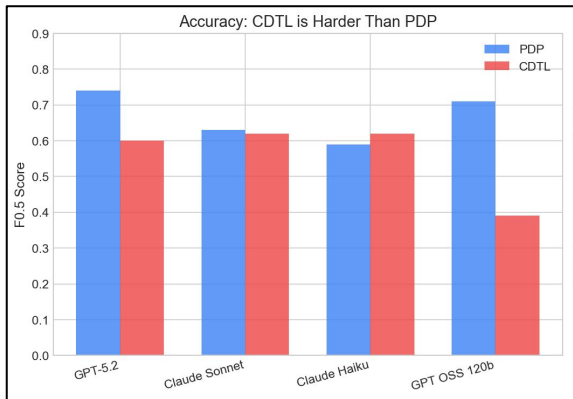
Results

Why Batch Mode Works Better

1. **Context awareness:** When the AI sees all fields together, it can maintain consistency. If it maps first_name to Person.Name.firstName, it's more likely to correctly map last_name to Person.Name.lastName rather than proposing an inconsistent target.
2. **Reduced prompt overhead:** In standard mode, the full LIF schema (521 fields) is sent with every request. In batch mode, it's sent once.
3. **Global reasoning:** The AI can recognize patterns across fields, such as address components that should all map to the same parent entity.



Results



Characteristic	PDP	CDTL
Best F0.5	0.74	0.62
Best Precision	71%	60%
Average Hallucinations	1.5 per run	24 per run

CDTL Field	AI Proposed	Why It's Wrong
Credential.audienceLevelType	Credential.level	"type of level" ≠ "level"
Course.audienceLevelType	Course.courseLevelType	Same semantic confusion
IdentifierValue.identifier	Person.Identifier.identifier	Nothing indicates "person"

Schema-Specific Findings

- CDTL proved significantly harder than PDP.
- **Vocabulary Overlap:** CDTL uses terms like “Competency,” “Credential,” “Assessment,” and “Course” extensively. These same terms appear throughout the LIF target schema. The AI finds many “close matches” that aren’t quite right
- **Higher Stakes:** There are more mappings to find but also more opportunities for false positives.

Source-to-target transformation



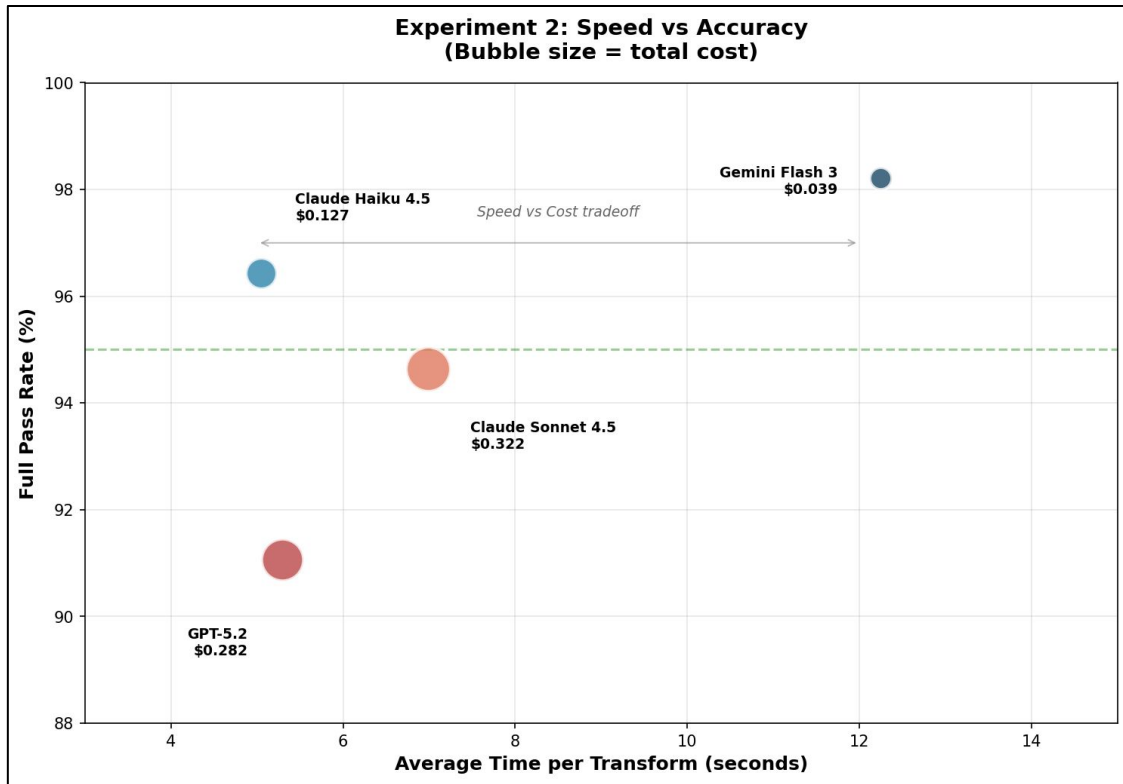
Results

Question: Can AI correctly identify when data values need conversion (e.g., date format changes, code translations)?

Model	Mode	PDP Accuracy	CDTL Accuracy	Notes
Gemini Flash 3	Batch	100%	86%	Perfect on PDP
Claude Sonnet 4.5	Batch	99%	86%	Excellent, no false positives
Claude Haiku 4.5	Batch	97%	84%	Strong performance
GPT-5.2	Standard	95%	83%	Works in standard mode
GPT-5.2	Batch	43%	75%	Severe over-prediction

Results

All tested models achieved strong results generating valid JSONata expressions, with full pass rates ranging from 91% to 98%. In a reversal of Experiment 1 findings, Tier 2 models outperformed Tier 3 models on this task.



Results



Results

1. **AI is a genuine capability leap:** Language models dramatically outperform traditional NLP methods
2. **Small models are cost-effective:** Small AI models nearly match large ones at a fraction of the cost
3. **Batch processing wins:** Processing all fields at once outperforms field-by-field approaches
4. **High accuracy for transformations:** AI generates valid transformation expressions with strong accuracy

Results

1. **AI cannot fully automate schema mapping.** Human experts must review proposals.
2. **AI changes the human task.** Instead of searching all field options for data mapping matches, humans verify a subset of data for training. This is faster but still requires expertise.
3. **False positives are the primary failure mode.** Our emphasis on precision (via F0.5) was appropriate; wrong mappings that look plausible can slip through review.

Results

1. **Rigorous comparison across tiers:** Traditional methods vs. small LLMs vs. large LLMs with controlled variables
2. **Batch mode discovery:** The finding that single-call batch processing outperforms field-by-field approaches appears to be novel
3. **Schema-specific reliability:** The finding that certain models fail specifically on technical vocabularies (CDTL) has practical implications
4. **Cost-accuracy tradeoffs:** Concrete data showing diminishing returns from model size, with specific recommendations

Where is DataKind taking this work
next?

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Agentic AI support on mapping (in experimentation)

MDR AI Mapping Agent

Standalone prototype of the AI-powered wire editor for the LIF Metadata Registry (MDR). This application automates schema field mapping and JSONata transform generation between source data models (PDP, CDTL) and the Learner Information Framework (LIF).

Built as a proof-of-concept for the [DataKind](#) ed-schema project. The LIF core team can port this directly into their platform; all outputs match the real MDR API contract.

Context: Where This Fits

The LIF data pipeline has six steps to onboard a new data source:

Step	Description	Who
1	Org decides to pull data from their SIS, LMS, or other system	Org
2	Someone defines a SourceSchema in MDR (entities + attributes)	Data engineer
3	Someone draws the wires (source field to target field) in the MDR UI	This agent
4	Someone writes/tweaks JSONata transform expressions	This agent
5	Someone creates an adapter and configures information_sources_config.yml	Developer
6	System runs: Adapter, Translator (uses the mappings), Cache	Automated

This agent replaces steps 3 and 4 with an AI-driven workflow: batch field mapping via a single LLM call, followed by per-field JSONata generation for fields requiring transformation.

Agentic AI support on mapping (in experimentation)



MDR AI Mapping Agent

Standalone prototype of the AI-powered wire editor for the LIF Metadata Registry (MDR). Automates schema field mapping and JSONata transform generation between source data models and LIF.

Mock MDR Import Mode ⓘ

Where this fits in the LIF data pipeline

1. Org decides to pull data from their SIS, LMS, or other system
2. Someone defines a SourceSchema in MDR (entities + attributes describing the external format)
- 3. Someone draws the wires (source field to target field) in the MDR UI**
- 4. Someone writes and tweaks JSONata transform expressions**
5. Someone creates an adapter and configures information_sources_config.yml
6. System runs: Adapter, Translator (uses the mappings), Cache

This agent automates steps 3 and 4 with AI.

Select source schema to map to LIF (157 target fields)

**PDP** 141 fields
Postsecondary Data Partnership
Flat schema with student enrollment, course, and outcome fields

**CDTL** 139 fields
Credential Transparency Description Language
Hierarchical schema with credential, assessment, and competency fields

Digital Holistic Student Supports (DHSS) Initiative

Over a two-year grant period (2026–2027), participating institutions will engage in co-designing and testing new approaches to improve data maturity and generate practical insights to inform the field.

- Co-designed, implementable solutions
- Readiness and adoption support
- Public good, community-engaged data model
- Knowledge publications
- Benchmark datasets

Data Mapper software

- Teams spend 100s of hours mapping data between systems with little data lineage tracking and version control, leading to brittle, low-trust data outputs
- Learning from our language model experimentation, built a software to guide mapping
- RAG-backed customized LLM model coupled with HITL can support high quality data mapping and documentation in a fraction of the time of common approaches

